How accurate are regression-based global treatment effect estimators under network interference? A simulation study of design, extrapolation and misspecification

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Abstract

Regression-based estimators of the global average treatment effect (GATE) fit an outcome model on exposure features from one randomized experiment and then predict outcomes under "everyone treated" and "everyone control". We ask how accurate this is, what determines the error, and when it fails. On synthetic outcomes over five graphs (SBM, Barabási–Albert, Watts–Strogatz, Facebook100 Stanford3, SNAP ego-Facebook), with Bernoulli and graph-cluster randomization, we report five findings, each supported by Monte-Carlo estimates with intervals. (i) When the regression is correctly specified, error is almost pure variance, scales like $n^{-1/2}$ and (in Bernoulli designs) like $\sqrt{d}$ in the degree d, and can be predicted from the realised design matrix alone (ratio of simulated to plug-in predicted SD: median 1.01, 10–90% range 0.94–1.27 across 128 configurations, with a rich model). (ii) Under misspecification the error is dominated by an extrapolation bias that does *not* shrink with n and grows linearly with interference strength; the key quantity is how far the observed treated-neighbour fraction lies from the all-treated point. (iii) Bernoulli designs observe neighbour exposures concentrated near p, so they extrapolate the most; cluster randomization with good clusters reduces both bias and variance by an order of magnitude, and the benefit degrades continuously as clusters leak edges. (iv) Naive heteroskedasticity-robust intervals attain nominal coverage only when heterogeneity covariates are included, otherwise coverage can fall to 17–43% on real graphs. (v) With very few clusters, or at extreme p, all estimators degrade, and a plug-in variance formula based on independent noise underestimates the error. We also report negative results: a Hajek exposure estimator is undefined on most Bernoulli draws; gradient boosting helps only when the true response saturates. Conclusions are limited to our synthetic one-hop interference models.

1 Introduction

Platform experiments rarely satisfy no-interference: a user's outcome depends on neighbours' assignments. The quantity of interest is usually the *global* average treatment effect, $\tau_{\mathrm{GATE}} = \frac{1}{n}\sum_i [Y_i(\mathbf{1}) - Y_i(\mathbf{0})]$. Difference in means (DM) is biased for it. Exposure-mapping and inverse-probability estimators [Aronow and Samii, 2017, Ugander et al., 2013] are unbiased under a correct exposure mapping, but they need overlap: units whose whole neighbourhood is treated (or controlled) must occur with non-negligible probability, which fails on dense graphs. Regression adjustment [Chin, 2019, Han and Ugander, 2023] instead *extrapolates*: fit $Y \sim f(z_i, \text{neighbour features})$ and

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predict at $\mathbf{1}$ and $\mathbf{0}$. It uses all units, but its accuracy rests on model validity far from the observed exposures. Existing work gives properties under linear models and resampling inference [Chin, 2019] and model/covariate selection [Han and Ugander, 2023]; we aim for an empirical map of *what determines its error*.

Contributions. (1) A controlled simulation study with known GATE across five outcome mechanisms, five graphs and two designs, with bias/variance decomposition (Sec. 4). (2) A simple plug-in prediction of the estimator's standard deviation from the realised design matrix, validated against simulation, including where it fails. (3) Evidence that the dominant risk is design-induced extrapolation, with sweeps over degree, treatment probability, cluster quality, interference strength, noise and sample size. (4) A coverage study of robust intervals. (5) Negative results. We do not propose a new estimator; we do not analyse real experiments.

2 Related work

Randomization-based inference under general interference with Horvitz–Thompson estimators is due to Aronow and Samii [2017]; graph-cluster randomization and network exposure to Ugander et al. [2013], with bias-reduction via design and neighbourhood-aware analysis studied by Eckles et al. [2017] and deployed at scale by Karrer et al. [2021]. Cortez et al. [2022] use staggered rollouts to avoid network knowledge under low-order interference. Regression-adjusted GATE estimation was proposed by Chin [2019] (linear models, resampling inference), and extended with model-free covariates and block bootstrap by Han and Ugander [2023]. Our study is complementary: it fixes simple, interpretable regression families and varies the *design and graph* to characterise error sources, rather than proposing new covariates.

3 Setting and methods

Graphs. A is an undirected unweighted graph on n nodes with degrees $d_i > 0$ (isolated nodes dropped; largest component for real graphs). Synthetic ($n{\approx}5000$): SBM (50 equal blocks, mean degree 20, fraction $\rho = 0.1$ of edges between blocks), Barabási–Albert ($m = 8$), Watts–Strogatz (degree 20, rewiring 0.1). Real: Facebook100 Stanford3 [Traud et al., 2012, Rossi and Ahmed, 2015] ($n = 11586$, 568,309 edges) and SNAP ego-Facebook [McAuley and Leskovec, 2012] ($n = 4039$, 88,234 edges). One graph realisation per synthetic family.

Designs. Bernoulli: $z_i \sim \mathrm{Bern}(p)$ i.i.d. Cluster: Louvain communities (planted blocks for SBM; 10 communities for Stanford3, 16 for ego-FB, many for BA/WS) receive i.i.d. $\mathrm{Bern}(p)$ assignments. Default $p = 0.5$.

Outcomes. $Y_i = 0.5\,\ell_i + \tau z_i + \gamma\, g(x_i, c_i, z_i) + \sigma\varepsilon_i$, with $\tau = \gamma = \sigma = 1$ by default, $\varepsilon_i \sim N(0,1)$, ℓ_i the standardised log-degree, $x_i = \frac{1}{d_i}\sum_j A_{ij} z_j$ the treated-neighbour fraction, $c_i = \sum_j A_{ij} z_j / d$. Mechanisms: `linfrac`: $g = x$; `lincount`: $g = c$; `quad`: $g = x^2$; `sigmoid`: g a logistic of $12(x - 0.5)$ rescaled to $[0,1]$ (saturating threshold); `inter`: $g = x(1 + z)$. The GATE is computed exactly from the noise-free model ($= 2$ except 3 for `inter`). All mechanisms are one-hop.

Estimators. DM. Regression estimators fit OLS of Y on features $\phi(z, x, c, \ell)$ and return $\hat\tau = \frac{1}{n}\sum_i[\hat f(\mathbf{1}) - \hat f(\mathbf{0})]$ where at $\mathbf{1}$: $z = 1, x = 1, c = d_i/\bar d$ and at $\mathbf{0}$ all zero. Feature sets: OLS-frac $[1, z, x]$; OLS-count $[1, z, c]$; OLS-rich $[1, z, x, c, zx, \ell]$; Poly $[1, z, x, x^2, zx, \ell]$; GBM: scikit-learn `HistGradientBoostingRegressor` on (z, x, c, ℓ) (150 iterations). For OLS-frac/count/rich/Poly we also form 95% intervals $\hat\tau \pm 1.96\,\widehat{\mathrm{se}}$ with the HC1 sandwich variance and the delta method (i.e. ignoring dependence between units through the graph). Hajek exposure estimator: treated-exposure = unit and all neighbours (closed neighbourhood) in treated clusters; weights are inverse exact exposure probabilities (p^{k_i}, k_i = number of clusters in the closed neighbourhood); undefined (NaN) if no unit is exposed.

Metrics. Bias, SD, RMSE of $\hat\tau - \tau_{\mathrm{GATE}}$ over $R = 300$ assignment/noise draws (60 for GBM) with 95% percentile bootstrap intervals for RMSE (200 resamples), Monte-Carlo SE of the bias, and coverage. We use RMSE since it combines the two error sources and bias/SD show which dominates. In absolute units, with $\tau_{\mathrm{GATE}} = 2$ in most cases.

Table 1: RMSE of the GATE estimate ($\tau_{\mathrm{GATE}} = 2$; $R = 300$, GBM 60). Full table (all mechanisms, estimators, intervals) in Appendix B. Smallest RMSE across rows is only meaningful within a row.

		`linfrac`				`sigmoid`			
Graph	Design	DM	Frac	Rich	GBM	DM	Frac	Rich	GBM
SBM	Bern	1.00	0.14	0.13	0.52	1.00	1.22	1.23	0.42
SBM	Clust	0.10	0.05	0.04	0.05	0.03	0.05	0.04	0.05
BA	Bern	1.00	0.11	0.10	0.26	1.00	0.94	0.94	0.43
BA	Clust	0.71	0.13	0.23	0.27	0.56	0.50	0.62	0.31
WS	Bern	1.00	0.15	0.13	0.49	1.00	1.26	1.26	0.45
WS	Clust	0.13	0.04	0.04	0.06	0.04	0.06	0.06	0.06
Stanford3	Bern	1.00	0.12	0.07	0.37	1.00	0.37	0.39	0.59
Stanford3	Clust	0.44	0.38	0.28	0.15	0.27	0.40	0.20	0.16
ego-FB	Bern	1.01	0.53	0.12	0.42	1.00	0.81	0.62	0.55
ego-FB	Clust	0.18	0.19	0.04	0.06	0.18	0.19	0.04	0.06

Plug-in variance heuristic. For correctly specified OLS with independent noise variance σ_e^2, $\mathrm{Var}(\hat{\tau}) = \sigma_e^2\, c^\top (X^\top X)^{-1} c$, where $c = \frac{1}{n}\sum_i [\phi(\mathbf{1}) - \phi(\mathbf{0})]$. We evaluate this on realised X (mean over 5 assignment draws). In a regular graph with Bernoulli design, $\mathrm{Var}(x_i) = p(1-p)/d$, so $\mathrm{SD}(\hat{\tau}) \approx \sigma_e \sqrt{(1+d)/(np(1-p))}$: the unit x moves only $\pm\sqrt{p(1-p)/d}$ around p in the data, yet we predict at $x = 1$, an extrapolation of $\sqrt{d(1-p)/p}$ standard deviations. This is a heuristic derivation, not a theorem; we test it.

4 Experiments

Code, seeds and commands are in the supplement. Seven experiments (E1–E7) vary graph×mechanism×design (E1), γ (E2), degree (E3), p (E4), SBM cluster leakage ρ (E5), n (E6), σ (E7).

E1: main comparison (Table 1). DM has RMSE ≈ 1 (bias $\approx \gamma$) under Bernoulli on every graph, i.e. it misses the whole interference effect. With a correct regression, RMSE is 0.04–0.15 on SBM/WS/ego-FB under both designs (cluster: ≈ 0.04), about a 7–8× gain over DM under Bernoulli assignment. With the saturating `sigmoid` mechanism and Bernoulli assignment, linear-in-x regressions are *worse than the truth by a bias of about* 1.2γ on SBM/WS (RMSE 1.2–1.3), and the richer models do not fix it, because the data contain almost no observations with x near 0 or 1. Under cluster designs, `sigmoid` errors drop to 0.04–0.06 on SBM/WS since many units have $x \approx 0$ or 1. On SBM/WS, DM is also close to unbiased under strong clustering with `sigmoid` (0.03–0.04), because treated units then really are surrounded by treated units: the regression gives no advantage there (0.04–0.06). The result is not universal: on BA with cluster design the regressions are *worse* (OLS-rich 0.22–0.38 for the non-sigmoid mechanisms and 0.62 for `sigmoid`; Poly up to 2.2). We did not isolate the cause; a plausible one is that hubs have neighbours in many clusters, so x is not extreme for the highest-leverage units. On Stanford3 (10 clusters) the cluster design is much *worse* than Bernoulli for `linfrac` (OLS-rich 0.28 vs. 0.07): with ≈10 randomisation units, the effective sample size is closer to the number of clusters than n (Sec. 4, E4).

E3/E6: variance of a correct model. For `linfrac` on WS-Bernoulli, SD of OLS-frac/rich scales as $n^{-0.53}$ and $n^{-0.52}$ over $n \in [10^3, 4\times10^4]$ (Fig. 2), and as $d^{0.46}$ and $d^{0.45}$ over $d \in [4, 128]$ at $n = 5000$: close to the $\sqrt{d/n}$ of the heuristic. The bias is within ± 0.025 (MC SE ≤ 0.02). Hence the estimator is accurate but its *sample size is effectively* n/d: at $d = 128$ RMSE is 0.32–0.37. In contrast, for `sigmoid` the RMSE is flat in n (1.28 → 1.24 for OLS-frac) since it is dominated by a constant bias (1.24–1.25); more data do not help.

Plug-in prediction (all E1–E5 configurations, 128 per estimator). The ratio simulated SD/predicted SD for OLS-rich has median 1.01 (10–90%: 0.94–1.27, max 5.4). For the correctly specified `linfrac` model on SBM and WS the ratio is 0.93–1.11 for OLS-frac and OLS-rich. The ratio is large where the iid-noise assumption breaks: configurations with few clusters (Stanford3: median ratio 3.2, up to 5.4; $p = 0.05$ with 50 clusters: ≈ 3.7), where ℓ_i-driven cluster-level residual variation and the small number of treated clusters add variance. OLS-frac omits ℓ whose variance (0.25) we add to σ_e^2; on

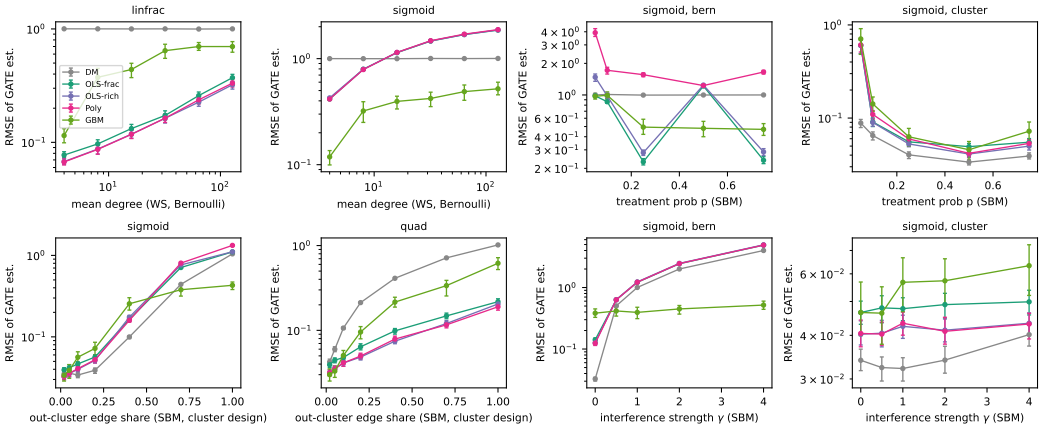


Figure 1: RMSE (log scale, 95% bootstrap intervals) of DM and regression estimators in sweeps. Top: mean degree (WS, Bernoulli; `linfrac`, `sigmoid`), treatment probability p on SBM (Bernoulli, cluster; `sigmoid`). Bottom: out-cluster edge share ρ of an SBM under cluster design (`sigmoid`, `quad`) and interference strength γ (`sigmoid`; Bernoulli, cluster).

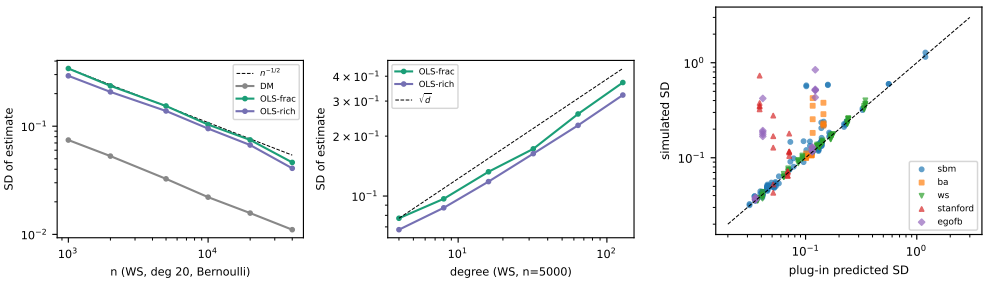


Figure 2: Left two: SD of the estimate vs. n and degree (`linfrac`, WS, Bernoulli). Right: simulated SD vs. plug-in predicted SD over all configurations in E1–E5 (OLS-frac and OLS-rich; points above the line are underestimated by the heuristic).

real graphs the ratio is then 4–6, so omitted covariates correlated with graph position are costly (see coverage below).

E2/E7: interference strength and noise. For correctly specified models the error is independent of γ (RMSE 0.12–0.15 across $\gamma \in [0, 4]$ for linfrac, Bernoulli), while under `sigmoid`+Bernoulli the bias is $0.615, 1.206, 2.433, 4.879$ for $\gamma = 0.5, 1, 2, 4$, i.e. $\approx 1.2\gamma$ (Fig. 1); with clusters it is ≤ 0.012 for all γ. Noise scaling is linear (E7, linfrac: RMSE 0.031/0.141/0.506 for OLS-rich at $\sigma = 0.25/1/4$).

E4: treatment probability. Under Bernoulli design the regressions have bias $-0.87, -0.77, -0.47, 0.00, +0.49$ (`quad`, OLS-frac, $p = 0.05, 0.10, 0.25, 0.5, 0.75$): the sign and size follow whether the observed $x \approx p$ lies on the same side of the curvature as the extrapolation targets; $p = 0.5$ is accidentally best for `quad` (the quadratic's secant slope between $x = 0.5$ and the targets happens to match the fitted slope) but worst for `sigmoid` (bias 1.21; at $p = 0.25$ only -0.15). That the best p flips sign across mechanisms shows that no p is safe under misspecification. For cluster designs and $p \in [0.1, 0.75]$ bias is ≤ 0.08 but at $p = 0.05$ (about 2.5 treated clusters of 50) SD is 0.57–0.58 and RMSE 0.60–0.63 (OLS-frac).

E5: cluster quality. On an SBM with cluster design, OLS-rich RMSE for `quad` increases smoothly from 0.032 ($\rho = 0.02$) to 0.205 ($\rho = 1$, i.e. clusters carry no information), and for `sigmoid` it is 0.033 at $\rho = 0.02$, 0.051 at 0.2, 0.17 at 0.4 and 1.1 at $\rho = 1$ with OLS-frac bias growing from 0.02 ($\rho = 0.2$) to 1.08. For the linear-type mechanism the loss is variance (SD $0.039 \to 0.218$); the nonlinear `sigmoid` adds bias once $\rho \gtrsim 0.4$. DM tracks the same pattern but is worse for `quad` throughout ($\rho = 0.1$: 0.106 vs. 0.041).

Interval coverage. For correctly specified models (`linfrac`, `inter`) OLS-rich and Poly with HC1 intervals cover at 0.93–0.97 on all graphs and designs. OLS-frac, which omits the log-degree covariate that enters the outcome, covers at 0.94–0.97 on SBM/WS, but at 0.39 (Bernoulli) and 0.31–0.32 (cluster) on ego-FB, 0.86–0.87 / 0.17–0.18 on Stanford3, and 0.59 on BA-`inter`-cluster. We hypothesise (not tested) that, because x_i and x_j share treated neighbours while the omitted component is fixed, the independence assumption in the sandwich fails exactly when the omitted heterogeneity varies with graph position.

Baselines. Gradient boosting is best for `sigmoid` under Bernoulli on SBM/WS (0.42/0.45 vs. 1.2–1.3), because trees extrapolate as constants and the true response saturates; but for linear-type mechanisms GBM has RMSE 0.26–0.8 under Bernoulli, 2.5–5$\times$ worse than OLS-rich, and only 60 repetitions were run. The Hajek exposure estimator is undefined in almost all Bernoulli draws on SBM, WS and Stanford3 (no unit has a fully treated and a fully control closed neighbourhood; shown as '–' in Appendix B), is undefined in 9–14% of draws on BA and well defined but noisy on ego-FB (RMSE 0.4–1.1, many weights near p^{d+1}). With clusters it is comparable to DM on SBM/WS (0.09–0.17) and about 2–3$\times$ worse than a correct regression; on BA with clusters it is undefined in up to 40% of draws.

5 Discussion and limitations

Answer to the question. Where interference is one-hop and the regression family contains the truth, regression-based GATE estimation is accurate (RMSE ≈ 0.04–0.15 on a GATE of 2), and its error is governed by a design-leverage term $c^\top (X^\top X)^{-1} c$ that we can compute *before* running the experiment. Where the family is wrong, the error is a bias that does not decay with n and scales with γ; the experiment's own exposure support determines its size, and cluster randomization with good clusters is the most effective mitigation we found. The advice implied by our results is conditional on this simulation: prefer designs whose exposure support covers 0 and 1; report the leverage $c^\top (X^\top X)^{-1} c$ and the exposure distribution; do not trust nominal intervals when heterogeneity covariates are omitted; treat the number of clusters, not n, as sample size.

Limitations. (1) Outcomes are synthetic, one-hop and the graph is known and exact; two-hop interference, mis-measured edges and temporal effects are untested. (2) One graph draw per synthetic family, so graph-to-graph variability is not in the intervals; our statements about BA vs. SBM are about one instance each. (3) The `sigmoid` failure is partly by construction: we chose a response whose extrapolation from the observed support is hard, and the `quad` bias sign depends on curvature; numbers like "1.2γ" are specific to this function. (4) SBM clusters use the planted partition (oracle) while real graphs use Louvain, giving fewer clusters (10, 16) than ideal and confounding "cluster quality" with "number of clusters". (5) We do not evaluate the resampling-based inference or covariate-selection procedures of the cited regression-adjustment papers, only HC1 delta-method intervals; the interval conclusions do not transfer to them. (6) The plug-in prediction is heuristic, fails with few clusters, and we did not derive a corrected formula. (7) The explanation for the coverage failure is a hypothesis. (8) GBM was run with default-ish hyper-parameters and 60 repetitions. (9) No outcome-noise dependence across neighbours or spillover heterogeneity beyond the stated mechanisms was studied. (10) A first pass with $R = 100$ (60→20 GBM reps) used for debugging gave a qualitatively identical picture and was overwritten by the reported $R = 300$ run; no configurations were dropped.

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A Reproduction details

Python 3.11, numpy 2.4.6, scipy 1.17.1, networkx 3.6.1, scikit-learn, pandas, matplotlib 3.11.2 (see `requirements.txt` in the code folder). Seeds: configuration k in `run_all.py` uses seed k for assignment and noise; synthetic graphs use seed 1; Louvain seed 0. Total compute: roughly 12 minutes on 36 processes (single-threaded BLAS), no GPU. Scripts: `run_all.py` $\rightarrow$ results JSON (E1–E7), `analyze.py` $\rightarrow$ Table B (`tab_main.tex`), Fig. 1, Fig. 2 (left) and coverage CSV, `maketab2.py` $\rightarrow$ Table 1, `predsd.py` $\rightarrow$ Fig. 2 (right) and the plug-in statistics. Data: Stanford3 zip sha256 `b1221e41...e8720d`, ego-Facebook sha256 `125e84db...35e`. Gradient-boosting settings: 150 iterations, learning rate 0.1. Bootstrap RMSE intervals: 200 resamples.

B Full RMSE table

Columns: DM, OLS-frac, OLS-count, OLS-rich, Poly, GBM, Hajek. Bold is the smallest in the row; †: estimator undefined in more than 5% of draws; '–': undefined in nearly all draws. `inter` has $\tau_{\mathrm{GATE}} = 3$.

Graph	Mech.	Design	DM	Frac	Count	Rich	Poly	GBM	Hajek
SBM	linfrac	Bern	1.00	0.14	1.76	**0.13**	0.13	0.52	–
SBM	linfrac	Clust	0.10	0.05	0.16	**0.04**	0.04	0.05	0.10
SBM	lincount	Bern	1.00	0.16	2.21	**0.12**	0.13	0.52	2.91†
SBM	lincount	Clust	0.11	0.05	0.25	**0.04**	0.04	0.05	0.11
SBM	quad	Bern	1.00	0.15	1.75	0.13	**0.12**	0.46	–
SBM	quad	Clust	0.11	0.05	0.16	**0.04**	0.04	0.04	0.10
SBM	sigmoid	Bern	1.00	1.22	2.44	1.23	1.22	**0.42**	–
SBM	sigmoid	Clust	**0.03**	0.05	0.24	0.04	0.04	0.05	0.10
SBM	inter	Bern	1.50	0.13	1.53	**0.12**	0.12	0.73	1.52†
SBM	inter	Clust	0.15	**0.04**	0.12	0.04	0.04	0.05	0.09
BA	linfrac	Bern	1.00	0.11	0.26	**0.10**	0.10	0.26	1.04†
BA	linfrac	Clust	0.71	**0.13**	0.64	0.23	0.87	0.27	0.65†
BA	lincount	Bern	1.00	0.30	0.69	**0.11**	0.30	0.34	1.16†
BA	lincount	Clust	0.74	0.34	0.78	**0.23**	1.05	0.36	0.80†
BA	quad	Bern	1.00	0.11	0.27	**0.10**	0.10	0.31	1.02†
BA	quad	Clust	0.72	0.35	0.50	0.38	2.22	**0.32**	0.62†
BA	sigmoid	Bern	1.00	0.94	**0.22**	0.94	0.94	0.43	1.03†
BA	sigmoid	Clust	0.56	0.50	0.39	0.62	2.47	**0.31**	0.62†
BA	inter	Bern	1.50	0.11	0.74	**0.10**	0.10	0.39	0.94†
BA	inter	Clust	1.09	0.25	0.63	**0.22**	0.94	0.38	0.67†
WS	linfrac	Bern	1.00	0.15	1.19	**0.13**	0.13	0.49	–
WS	linfrac	Clust	0.13	**0.04**	0.26	0.04	0.04	0.06	0.13
WS	lincount	Bern	1.00	0.15	1.26	**0.13**	0.13	0.51	–
WS	lincount	Clust	0.14	**0.04**	0.29	0.04	0.04	0.07	0.15
WS	quad	Bern	1.00	0.14	1.18	**0.13**	0.13	0.52	–
WS	quad	Clust	0.14	**0.04**	0.25	0.04	0.04	0.06	0.13
WS	sigmoid	Bern	1.00	1.26	2.32	1.26	1.27	**0.45**	–
WS	sigmoid	Clust	**0.04**	0.06	0.31	0.06	0.05	0.06	0.15
WS	inter	Bern	1.50	0.14	1.14	**0.13**	0.13	0.68	–
WS	inter	Clust	0.20	**0.04**	0.25	0.04	0.04	0.05	0.13
Stanford3	linfrac	Bern	1.00	0.12	0.21	**0.07**	0.07	0.37	–
Stanford3	linfrac	Clust	0.44	0.38	0.47	0.28	**0.13**	0.15	0.43
Stanford3	lincount	Bern	1.00	0.92	0.78	**0.06**	0.89	0.29	–
Stanford3	lincount	Clust	0.64	0.87	0.42	**0.17**	0.58	0.33	1.03
Stanford3	quad	Bern	1.00	0.10	0.25	0.09	**0.06**	0.25	–
Stanford3	quad	Clust	0.46	0.35	0.17	0.13	**0.12**	0.13	0.43
Stanford3	sigmoid	Bern	1.00	0.37	**0.20**	0.39	0.36	0.59	–
Stanford3	sigmoid	Clust	0.27	0.40	0.21	0.20	**0.16**	0.16	0.43
Stanford3	inter	Bern	1.50	0.11	0.71	**0.06**	0.06	0.30	–
Stanford3	inter	Clust	0.68	0.32	0.33	**0.04**	0.04	0.15	0.47
ego-FB	linfrac	Bern	1.01	0.53	0.35	**0.12**	0.12	0.42	0.65
ego-FB	linfrac	Clust	0.18	0.19	0.11	**0.04**	0.04	0.06	0.28
ego-FB	lincount	Bern	1.00	1.13	0.65	**0.13**	0.77	0.38	1.10
ego-FB	lincount	Clust	0.40	0.42	0.10	**0.04**	0.19	0.06	0.64
ego-FB	quad	Bern	1.00	0.43	0.36	0.14	**0.11**	0.38	0.63
ego-FB	quad	Clust	0.17	0.17	0.10	**0.04**	0.04	0.05	0.27
ego-FB	sigmoid	Bern	1.00	0.81	**0.32**	0.62	0.63	0.55	0.65
ego-FB	sigmoid	Clust	0.18	0.19	0.11	**0.04**	0.04	0.06	0.30
ego-FB	inter	Bern	1.50	0.51	0.84	**0.11**	0.12	0.33	0.64
ego-FB	inter	Clust	0.18	0.18	0.10	**0.04**	0.04	0.05	0.27

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